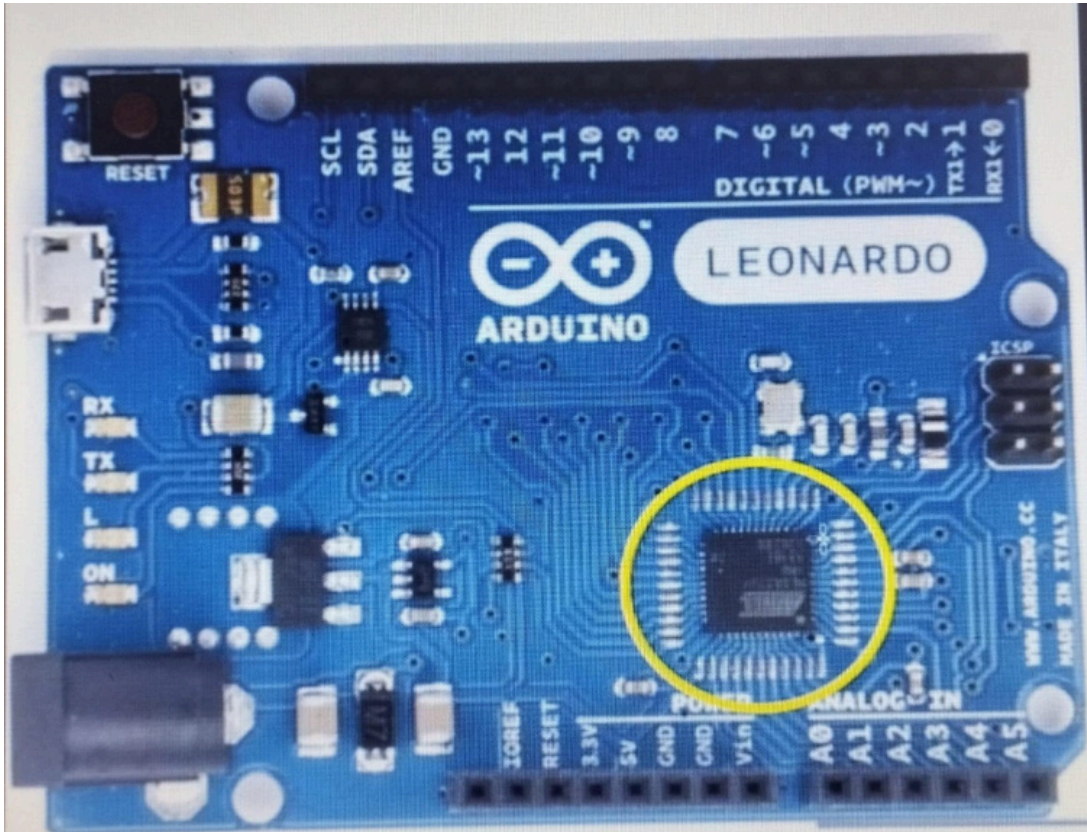


How to talk to computers

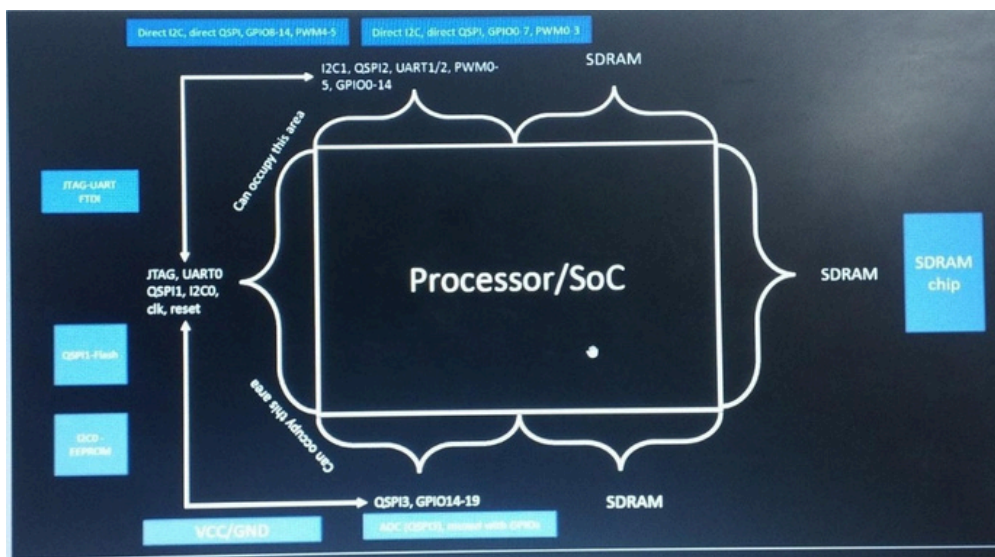
Electronic circuit or electronic board

Eg. Arduino board



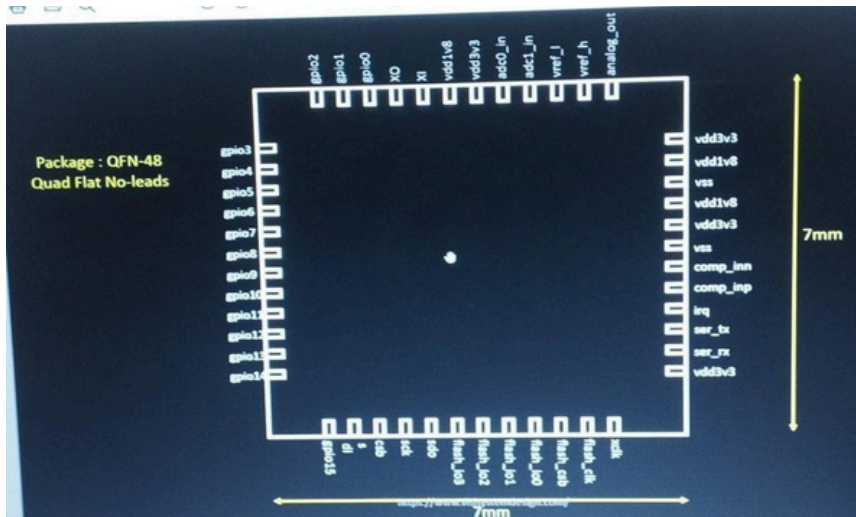
Processor is circled.

Block diagram of the above board



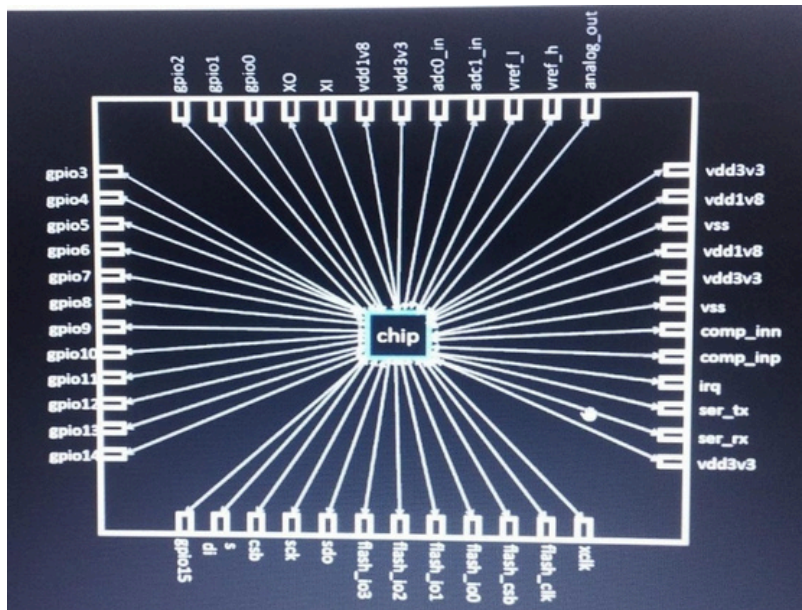
CHIP DESIGNING COURSE

While opening the above Processor IC



Its technically called PACKAGE. The above package is called QFN-48 (QUAD FLAT NO LEADS) and it is 7mm*7mm dimension. There are multiple such packages in the market. The pin locations of the package are driven by the board design.

Chip is normally placed at the centre and is connected to the pins.



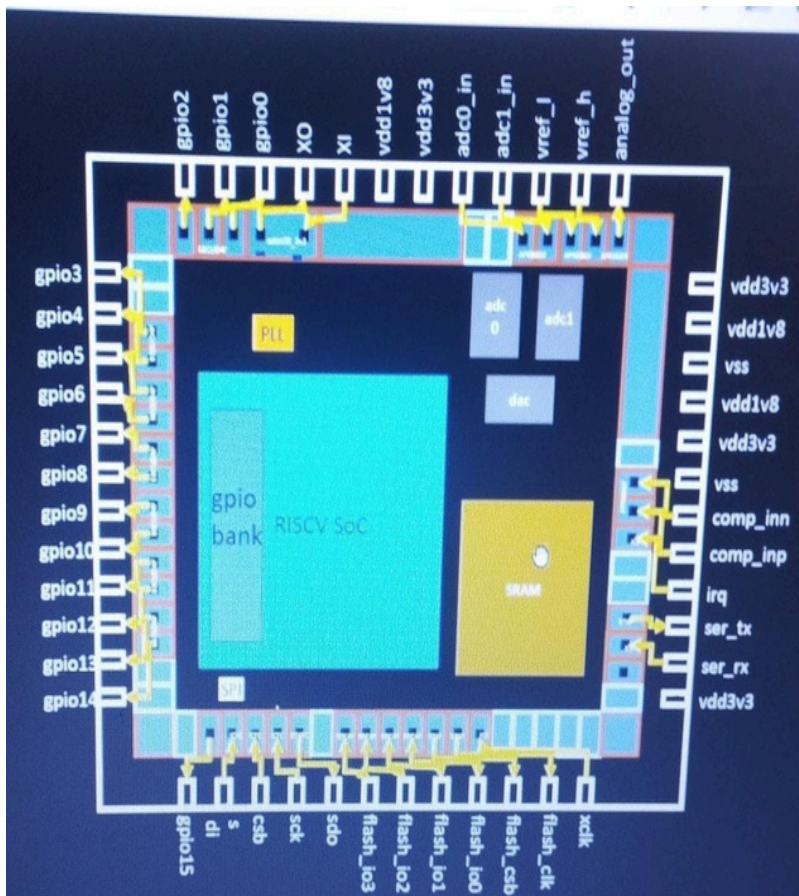
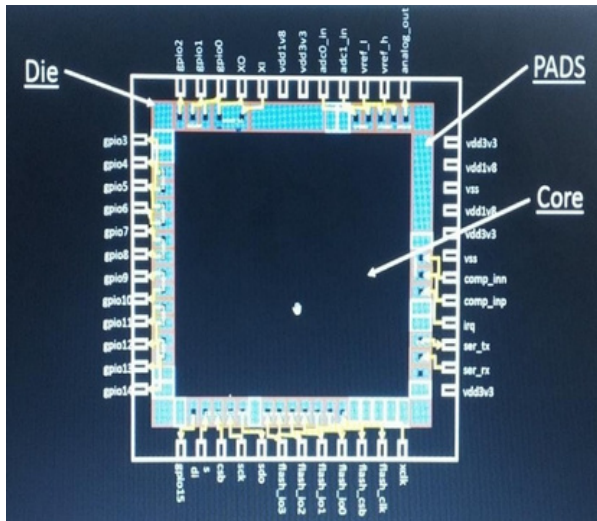
While opening the chip - its components

PADS - send signals inside / outside the chip

CORE - where digital logical gates are placed (AND.OR etc)

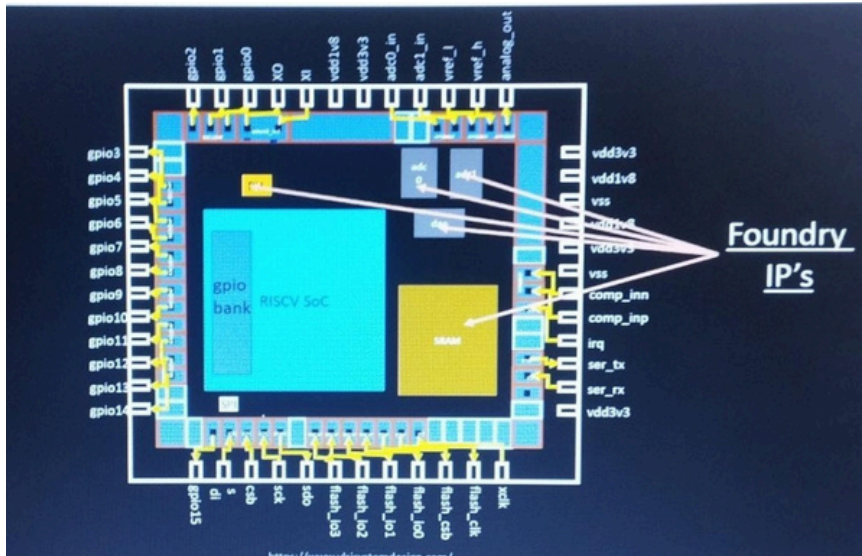
Die - size of the entire chip made up of silicon

CHIP DESIGNING COURSE

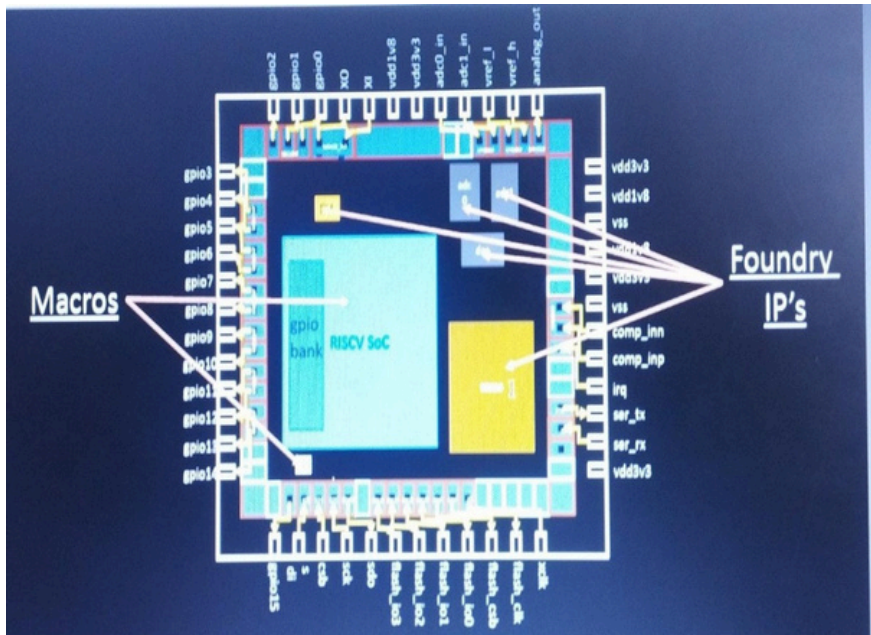


In the above RISC-V SOC

Core consists of RISC-V, SRAM, SPI, PLL, ADC, DAC



Foundry is a factory where chips are manufactured.



FOUNDY IPs -needs intelligence - Intellectual property

MACROS - pure digital logic

To build the chip foundry is the one to be communicated ..to Communicate with foundries interface files are needed.

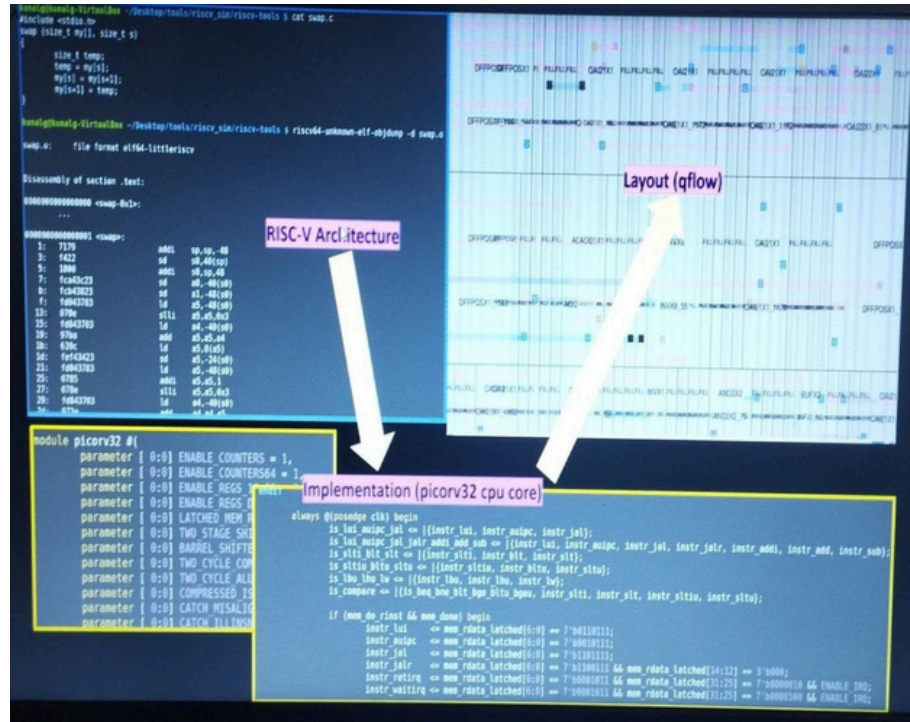
RISC-V Instruction set architecture

Language of the computer - to communicate with the computer

If C prog has to be run on a hardware a certain instruction has to be passed in certain terms

CHIP DESIGNING COURSE

C prog is in assembly language which has to be converted into machine language prog (binary)and when it gets executed in the layout and req output is got. Another interface has to be present between RISK V and the layout is the hardware description language (HDL)...In our prjct we are taking picorv32 cpu core.



How does application software (APPS) run on hardware ?

Application software -> System software -> Binary language

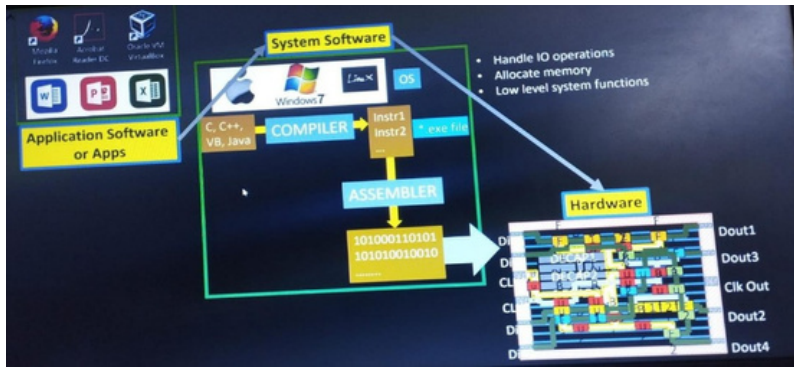
System software components

OS memory allocation ,software to respective assembly language and binary lang
its out put is in C/C++

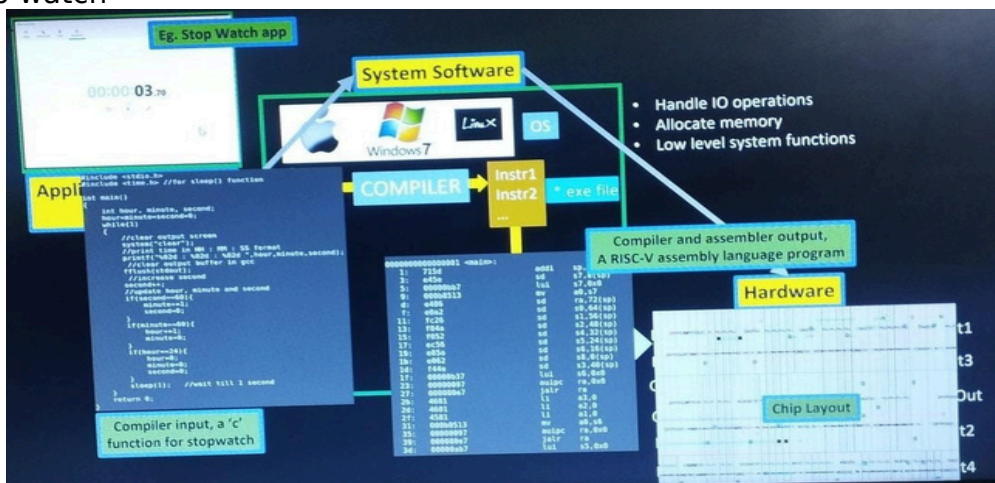
COMPILER -Respective compiler takes the C/C++ and gives instructions ..files will have .exe

ASSEMBLER to take the instructions and convert it into respective binary language
- called as machine language prog- language the machine understands -
Gives the result to the hardware which follows them and gives result.

CHIP DESIGNING COURSE

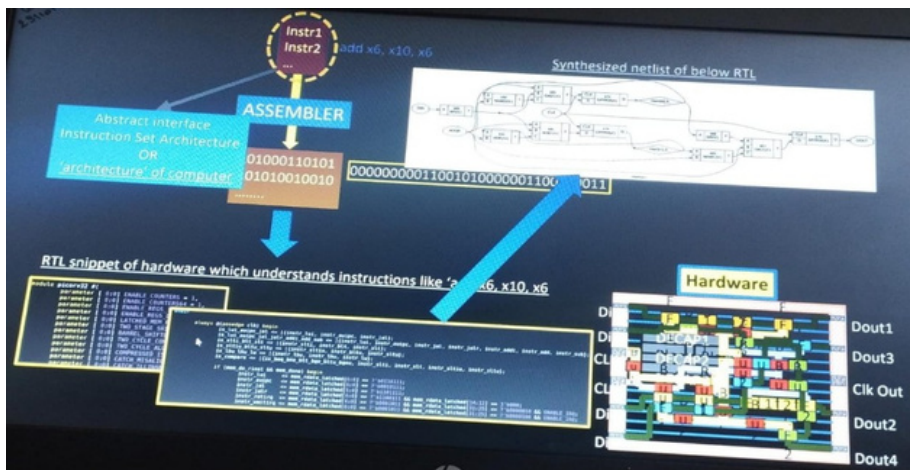


Eg
Stop watch



In the course we will be concentrating on RISC V architecture.

Eg.



CHIP DESIGNING COURSE

For the command to add x6,x10,x6 ...the assembler result and the binary out put are given .Hardware description language -Which in turn RTL is synthesized in netlist as in the above pic.

Then Physical design implementation of the above netlist happens.